

COMPUTER NETWORKS

Complete Technical Course Handbook & Detailed Syllabus Notes

Teaching Scheme: 3-0-0 (L-T-P)

Coverage: Units I through V (End-to-End)

Focus: Protocols, Architecture, & Layers

UNIT I: INTRODUCTORY CONCEPTS & PHYSICAL LAYER

1. Introduction to Computer Networks

A **Computer Network** is a set of interconnected autonomous computing devices designed to share resources, information, and services. The fundamental goals of computer networks include resource sharing, high reliability (via redundancy), cost reduction, and providing a powerful communication medium.

Categories of Networks & Internet Organization

- **LAN (Local Area Network):** Covers a small geographic area (office, building). High speed, low error rate.
- **MAN (Metropolitan Area Network):** Covers a city. Typically used for connecting multiple LANs or cable television networks.
- **WAN (Wide Area Network):** Covers large geographical areas (countries, continents). Uses routers and public communication links.
- **The Internet & ISPs:** The Internet is a "network of networks." It is organized hierarchically via **Internet Service Providers (ISPs)**. Tier-1 ISPs form the internet backbone (international coverage), Tier-2 ISPs provide regional coverage, and Tier-3 ISPs provide last-mile connectivity to end-users.

2. Network Architecture & Reference Models

Network architecture relies on **Layering Principles** to break down complex communication processes into smaller, manageable sub-tasks. Each layer provides a service to the layer above it and utilizes the services of the layer below it, governed by established **Protocols** (rules for data exchange) and **Standards**.

The OSI Reference Model

The Open Systems Interconnection (OSI) model defines a 7-layer theoretical framework for network communication:

Layer	Name	Primary Function & Protocols
Layer 7	Application	Network process to application. Provides user interfaces and services. (HTTP, FTP, SMTP, DNS)

Layer	Name	Primary Function & Protocols
Layer 6	Presentation	Data representation, syntax, encryption, and compression. (SSL/TLS, JPEG, ASCII)
Layer 5	Session	Interhost communication. Establishes, maintains, and terminates sessions. (NetBIOS, RPC)
Layer 4	Transport	End-to-end connections, reliability, flow control, and multiplexing. (TCP, UDP)
Layer 3	Network	Logical addressing (IP) and routing data across multiple networks. (IPv4, IPv6, ICMP, IPsec)
Layer 2	Data Link	Node-to-node data transfer, framing, MAC addressing, and error detection. (Ethernet, 802.11 MAC)
Layer 1	Physical	Transmission of raw bit streams over a physical medium. (Cables, Hubs, Repeaters)

TCP/IP Protocol Suite

The pragmatic model driving the modern Internet. It condenses the OSI model into 4 layers: **Application** (OSI 5, 6, 7), **Transport** (OSI 4), **Internet** (OSI 3), and **Network Access / Link** (OSI 1, 2).

3. The Physical Layer

The Physical Layer is responsible for moving individual bits from one node to the next. It deals with the mechanical and electrical specifications of the interface and transmission medium.

Network Topology Design

- **Mesh:** Every node is connected to every other node. Number of links = $n(n-1)/2$. Highly reliable but expensive.
- **Star:** All devices connected to a central hub/switch. Easy to install, but central node failure brings down the network.
- **Bus:** All devices share a single communication cable (backbone). Easy to extend, but prone to collisions and single-point cable failure.
- **Ring:** Devices are connected in a closed-loop. Data travels in one direction.

Transmission Media & Signal Encoding

Guided Media: Twisted Pair (UTP/STP), Coaxial Cable, Fiber Optic Cable (uses total internal reflection).

Unguided Media: Wireless (Radio waves, Microwaves, Infrared).

Signal Encoding Techniques

- **Line Coding:** Converts digital data to digital signals (e.g., NRZ, Manchester, Differential Manchester). Manchester encoding provides self-clocking by ensuring a transition in the middle of every bit.
- **Modulation:** Converts digital data to analog signals for transmission over analog media (e.g., ASK, FSK, PSK, QAM).

Network Performance & Transmission Impairments

Signals degrade as they travel. Key impairments include **Attenuation** (loss of energy), **Distortion** (alteration of signal shape), and **Noise** (thermal noise, crosstalk, impulse noise).

Nyquist Bit Rate (Noiseless Channel): $Capacity = 2 \cdot Bandwidth \cdot \log_2(L)$

Shannon Capacity (Noisy Channel): $Capacity = Bandwidth \cdot \log_2(1 + SNR)$

Switching Techniques & Multiplexing

- **Multiplexing:** Transmitting multiple signals over a single data link.
 - *FDM (Frequency Division Multiplexing):* Assigns different frequency bands to different signals (Analog).
 - *TDM (Time Division Multiplexing):* Assigns different time slots to different signals (Digital).
 - *WDM (Wavelength Division Multiplexing):* FDM applied to optical fiber using different colors of light.
- **Switching:**
 - *Circuit Switching:* Dedicated physical path established for the duration of the communication (e.g., legacy telephone networks).
 - *Packet Switching:* Data is broken into packets. Each packet routes independently. Highly efficient for bursty computer traffic (e.g., the Internet).

UNIT II: DATA LINK LAYER & MEDIUM ACCESS CONTROL

1. The Data Link Layer: Framing & Error Control

The Data Link Layer (DLL) transforms the physical layer into a reliable link. It is typically divided into two sublayers: Logical Link Control (LLC) and Media Access Control (MAC).

Framing

Framing encapsulates network layer packets into frames by adding a header and trailer. To ensure the receiver can identify frame boundaries, protocols use techniques like:

- **Byte Stuffing:** Adding a special escape byte (e.g., `ESC`) before any accidental occurrence of the flag byte in the payload.

- **Bit Stuffing:** Used in HDLC. If five consecutive `1`s appear in the payload, a `0` is automatically stuffed to prevent accidental generation of the flag sequence `01111110`.

Error Detection and Correction

Due to physical impairments, bits can flip. Error control mechanisms include:

- **Parity Check:** Adds a single bit to make the number of 1s even (or odd). Detects single-bit errors.
- **Cyclic Redundancy Check (CRC):** A powerful polynomial-based mechanism. The sender divides the data by a predetermined generator polynomial. The remainder is appended to the frame. The receiver performs the same division; a non-zero remainder indicates an error.
- **Hamming Code (Correction):** Uses multiple parity bits positioned at powers of 2. Can detect and *correct* single-bit errors.

Condition for redundant bits r given m data bits: $2^r \geq m + r + 1$.

2. Flow Control & Elementary Data Link Protocols

Flow control prevents a fast sender from overwhelming a slow receiver. Protocols include:

Sliding Window Protocols

- **Stop-and-Wait ARQ:** Sender transmits one frame and waits for an ACK before sending the next. Highly inefficient on links with high propagation delay (high bandwidth-delay product).
- **Go-Back-N ARQ:** Sender can transmit up to N frames without receiving an ACK. If a timeout occurs or a NAK is received for frame i , the sender retransmits frame i and *all subsequent frames*. Window size must be $\leq 2^m - 1$.
- **Selective Repeat ARQ:** Only the specific damaged or lost frame is retransmitted. The receiver must buffer out-of-order frames. Window size must be $\leq 2^{m-1}$.

3. Medium Access Control (MAC) & LAN Standards

When multiple stations share a single broadcast link, a MAC protocol is required to coordinate access and avoid collisions.

Multiple Access Protocols

- **Random Access (Contention-based):**
 - **ALOHA:** Stations transmit whenever they have data. If a collision occurs, they wait a random time and retry. Maximum efficiency is 18.4% (Pure) or 36.8% (Slotted).
 - **CSMA/CD (Carrier Sense Multiple Access with Collision Detection):** Used in traditional Ethernet. "Listen before you talk." If a collision is detected while transmitting, the station sends a jam signal and backs off for a random time calculated via the Binary Exponential Backoff algorithm.
 - **CSMA/CA (Collision Avoidance):** Used in Wireless LANs (802.11), since detecting collisions on a radio antenna is difficult. Uses RTS/CTS (Request to Send / Clear to Send) control frames to reserve the channel.
- **Controlled Access:** Token Passing (Token Ring, FDDI). A station can only transmit if it possesses the token.
- **Channelization:** FDMA, TDMA, CDMA.

LAN Standards (IEEE 802) & Ethernet

Ethernet (IEEE 802.3) is the dominant LAN standard. It defines a standard frame format containing Destination MAC (6 bytes), Source MAC (6 bytes), EtherType (2 bytes), Payload (46-1500 bytes), and FCS/CRC (4 bytes).

4. Link Layer Switches & Bridges

Switches and Bridges operate at Layer 2. They filter and forward frames based on MAC addresses.

- **Learning Bridge:** Builds a MAC forwarding table dynamically. When a frame arrives, it associates the Source MAC with the incoming port. It then looks up the Destination MAC; if known, it forwards to the specific port. If unknown, it *floods* the frame out of all other ports.
- **Spanning Tree Protocol (STP) [IEEE 802.1D]:** In networks with redundant switches, broadcast storms can occur due to loops. STP solves this by logically blocking certain ports to create a loop-free logical tree structure.
Steps: 1. Elect a Root Bridge (lowest bridge ID). 2. Determine Root Ports on other bridges (lowest path cost to root). 3. Determine Designated Ports for each LAN segment. 4. Block all other ports.

UNIT III: NETWORK LAYER & INTERNETWORKING

1. Network Layer Principles & Logical Addressing

The Network Layer is responsible for routing packets from the source host to the destination host across multiple networks, often involving varying protocols (Internetworking).

Logical Addressing (IPv4)

An IPv4 address is a 32-bit logical address. Originally, addresses were divided into Classes (A, B, C, D, E), but this led to massive address exhaustion.

Classless Inter-Domain Routing (CIDR)

CIDR eliminates class boundaries. An IP address is accompanied by a prefix length (e.g., `192.168.1.0/24`), indicating the number of bits used for the Network ID.

- **Subnetting:** Borrowing host bits to create smaller logical networks. Helps limit broadcast domains and manage IP allocation efficiently.
- **Subnet Mask:** A 32-bit mask used to mathematically extract the network address from an IP address via a bitwise AND operation.

2. Basic Internetworking Protocols

- **IP (Internet Protocol):** A connectionless, best-effort delivery protocol. The IPv4 header includes fields like TTL (Time to Live, prevents routing loops), Protocol (identifies the Layer 4 payload), and Fragmentation offsets (for breaking large packets to fit MTU limits).

- **ARP (Address Resolution Protocol):** Maps a logical IP address to a physical MAC address on a local network segment.
- **RARP/DHCP:** DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses, subnet masks, and default gateways to hosts dynamically. (Replaced the legacy RARP).
- **ICMP (Internet Control Message Protocol):** Used for network diagnostics and error reporting. Utilities like `ping` (Echo Request/Reply) and `traceroute` rely on ICMP.

3. Routing, Forwarding & Delivery

Forwarding is the local action of moving an arriving packet from an input link to the appropriate output link inside a router. **Routing** is the network-wide process of determining the end-to-end paths packets take from source to destination.

Routing Algorithms and Protocols

Algorithm Type	Mechanism	Protocols & Characteristics
Distance Vector (DV)	Routers share their entire routing table only with their immediate neighbors at regular intervals (Bellman-Ford algorithm).	RIP, EIGRP. Suffers from the "Count-to-Infinity" problem, mitigated by Split Horizon and Poison Reverse.
Link State (LS)	Routers flood information about their immediate links to the entire network. Each router builds a complete topology map (Dijkstra's Shortest Path algorithm).	OSPF, IS-IS. Faster convergence, requires more CPU/memory, highly scalable via hierarchical areas.
Path Vector	Exchanges AS (Autonomous System) path information to prevent routing loops across the global internet.	BGP (Border Gateway Protocol). The protocol that runs the Internet backbone.

4. Congestion Control & IPv6

Congestion occurs when the load on the network exceeds its capacity. The Network Layer implements mechanisms to handle this:

- **Traffic Shaping/Policing:** E.g., Leaky Bucket algorithm (enforces a constant output rate) and Token Bucket algorithm (allows bursty traffic up to a limit).
- **Choke Packets:** A router sends a warning packet back to the source host to reduce its transmission rate.
- **Random Early Detection (RED):** Routers proactively drop packets before the queue is completely full, signaling TCP senders to slow down.

IPv6 (Internet Protocol Version 6)

Designed to solve IPv4 address exhaustion. Key features:

- **128-bit Address Space:** Hexadecimal representation (e.g., `2001:0db8:85a3::8a2e:0370:7334`).
- **Simplified Header:** Removed fragmentation fields (handled by source) and header checksum (handled by L2/L4) for faster processing.

- **Built-in Security:** IPsec is natively integrated.

UNIT IV: TRANSPORT LAYER

1. Process-to-Process Delivery & Transport Protocols

While the Network Layer delivers packets from host to host, the Transport Layer delivers messages from a specific *process* on one host to a specific *process* on another host. This is achieved using **Ports** (16-bit addresses, e.g., HTTP on port 80). This multiplexing allows multiple applications to communicate simultaneously over a single IP connection.

UDP (User Datagram Protocol)

A connectionless, unreliable protocol. It adds only a minimal 8-byte header (Source Port, Dest Port, Length, Checksum) over IP. It does not provide flow control, error recovery, or sequencing. Used for real-time applications where low latency is prioritized over absolute reliability (e.g., VoIP, Video Streaming, DNS).

TCP (Transmission Control Protocol)

A connection-oriented, reliable, byte-stream protocol. TCP guarantees in-order, error-free delivery of data.

- **Connection Management:** Uses a **3-Way Handshake** to establish a connection.

1. Client sends **SYN** (Synchronize sequence numbers).
2. Server responds with **SYN-ACK**.
3. Client replies with **ACK**.

Connection termination uses a 4-way handshake (FIN, ACK, FIN, ACK).

- **Flow Control & Window Management:** TCP uses a dynamic Sliding Window protocol. The receiver advertises its **Receive Window (rwnd)** size in every ACK, telling the sender exactly how many bytes it can accept before its buffer fills up.

2. Flow Control, Retransmission, and TCP Congestion Control

TCP handles lost packets via timeouts and duplicate acknowledgments. If an ACK is not received within the Retransmission Timeout (RTO) interval, the packet is retransmitted.

TCP Congestion Control Phases

TCP maintains a **Congestion Window (cwnd)**. The actual sending window is the minimum of **cwnd** and **rwnd**.

1. **Slow Start:** Starts with **cwnd = 1 MSS** (Maximum Segment Size). **cwnd** doubles every RTT (exponential growth) until it reaches the Slow Start Threshold (ssthresh).
2. **Congestion Avoidance (AIMD):** Once **ssthresh** is reached, **cwnd** grows linearly (adds 1 MSS per RTT) to probe for bandwidth gently (Additive Increase).
3. **Congestion Detection:**
 - *Timeout (Severe Congestion):* **ssthresh** is set to **cwnd / 2**, and **cwnd** drops aggressively back to 1 MSS. Slow start resumes.
 - *3 Duplicate ACKs (Mild Congestion):* Triggers **Fast Retransmit** (retransmits the missing segment immediately without waiting for timeout) and **Fast Recovery** (halves **cwnd** and enters Congestion Avoidance rather than Slow Start) (Multiplicative Decrease).

3. Quality of Service (QoS)

QoS refers to the ability of a network to provide better service to selected network traffic over various technologies. Key parameters include:

- **Bandwidth:** Required data rate.
- **Delay/Latency:** Time taken for data to travel end-to-end.
- **Jitter:** Variation in packet delay (critical for voice/video).
- **Reliability:** Acceptable packet loss rate.

QoS mechanisms include IntServ (guaranteed per-flow reservations using RSVP) and DiffServ (classifying packets into differentiated traffic classes).

UNIT V: APPLICATION LAYER & CRYPTOGRAPHY

1. Domain Name System (DNS)

DNS acts as the phonebook of the Internet. It translates human-readable domain names (e.g., **www.google.com**) into machine-readable IP addresses (e.g., **142.250.190.4**).

- **Hierarchical Structure:** Root servers (.), Top-Level Domain servers (.com, .org), and Authoritative Name Servers.
- **Resolution Process:** Can be *Iterative* (client queries servers one by one) or *Recursive* (servers query other servers on behalf of the client).

2. Web, Email, and File Transfer Protocols

- **HTTP (HyperText Transfer Protocol):** The foundation of the WWW. Operates on TCP port 80. Uses Request/Response messages.
 - *Methods:* GET, POST, PUT, DELETE.
 - *Status Codes:* 200 (OK), 404 (Not Found), 500 (Internal Server Error).
 - *Stateless:* HTTP does not remember past requests; cookies are used to maintain sessions.
- **Electronic Mail (Email):**
 - *SMTP (Simple Mail Transfer Protocol):* Push protocol used to send mail from client to server, and server to server (Port 25).
 - *POP3 / IMAP:* Pull protocols used by clients to retrieve mail from the server. IMAP allows synchronization across multiple devices, whereas POP3 typically downloads and deletes.
- **FTP (File Transfer Protocol):** Used for reliable file transfer. Uniquely uses two parallel TCP connections: a Control Connection (Port 21) for commands and a Data Connection (Port 20) for actual file transfer.
- **Remote Login:** *Telnet* (legacy, unencrypted) and *SSH* (Secure Shell, encrypted on Port 22) allow remote command-line access to servers.

3. Network Management & Data Compression

- **SNMP (Simple Network Management Protocol):** Used for monitoring and managing devices on IP networks (routers, switches). Consists of a Manager, Agents, and MIBs (Management Information Bases).
- **Data Compression:** Reduces payload size to save bandwidth. Can be *Lossless* (text compression like Huffman coding, LZW) or *Lossy* (multimedia compression like JPEG, MPEG where slight quality degradation is acceptable).

4. Cryptography: Basic Concepts

Cryptography ensures network security, focusing on Confidentiality, Integrity, Authentication, and Non-repudiation.

Cryptography Type	Mechanism	Examples & Use Cases
Symmetric Key (Secret Key)	The same key is used for both encryption and decryption. Very fast, but key distribution over an insecure network is a major challenge.	AES, DES, RC4. Used for bulk data encryption (e.g., file encryption, Wi-Fi WPA2).
Asymmetric Key (Public Key)	Uses a mathematically linked key pair: a Public Key (shared openly) and a Private Key (kept secret). Data encrypted with the public key can only be decrypted by the private key.	RSA, Elliptic Curve (ECC). Used for secure key exchange, digital signatures, and establishing HTTPS (SSL/TLS) sessions.
Cryptographic Hash Functions	One-way mathematical functions that map variable-length data to a fixed-length string (hash or digest).	SHA-256, MD5. Used for verifying data integrity and storing passwords.